

Program Designing 'and' { [Programming].ICT 1305

< J.M.D.S.Jayasinghe >

< Department of ICT >

< Faculty of Technology >

< Rajarata University of Sri Lanka >

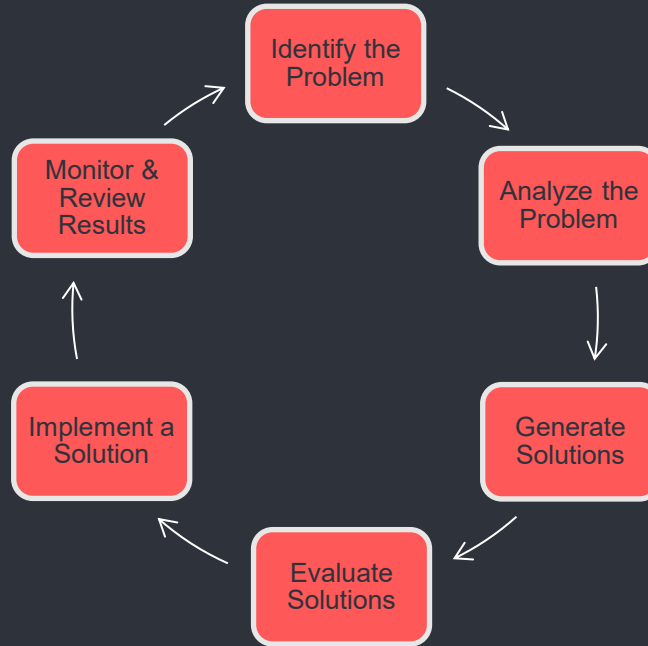
}

Problem Solving {

- Problem-solving is the process of identifying a challenge or issue, understanding its nature, and devising and implementing an effective solution.
- It is a critical skill used in various aspects of life, including personal decision-making, professional settings, and academic pursuits.

}

Phases of Problem Solving {



}

Phases of Problem Solving {

■ Identify the Problem

- Clearly identify and understand the problem.
- Gather necessary data and feedback.

■ Analyze the Problem

- Investigate the underlying root cause of the problem.
- Use tools like 5 Whys or Fishbone Diagram to analyze the root cause.
- Break the problem into smaller, manageable components.

}

Phases of Problem Solving {

■ Generate Solutions

- Generate a variety of possible solutions.
- Encourage creative thinking for the same.

■ Evaluate Solutions

- Critically examine each possible solution.
- Choose the most practical solution

}

Phases of Problem Solving {

■ Implement a Solution

- Evaluate all the possible solutions.
- Select the most feasible solution.
- Execute it with a structured plan.

■ Monitor & Review the Result

- Track the effectiveness of the solution.
- Make changes if necessary to ensure success.

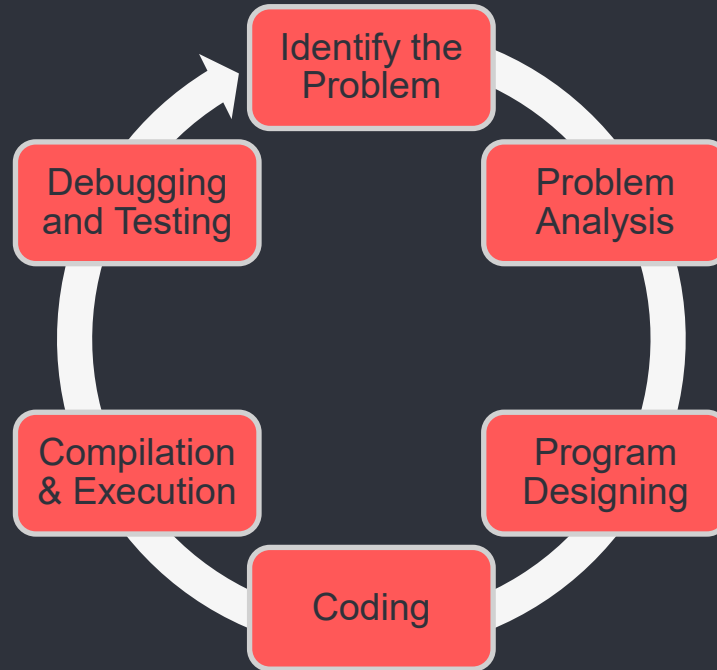
}

Problem Solving using Computer {

- Problem solving using computers refers to utilizing computational tools, methods, and programming techniques to analyze, design, and implement solutions for various problems.
- This process combines human logic with the computational power of machines to tackle complex tasks effectively and efficiently.
- After its creation, the software should be error free and well documented.
- Software development is the process of creating such software, which satisfies end user's requirements and needs.

}

Phases of Problem Solving using Computer {



Identify the Problem {

- In this step, the problem is clearly recognized and defined.
- The programmer must understand what needs to be solved and why a computer solution is required.
- Identifying the problem correctly is very important because an incorrect understanding can lead to an incorrect solution.
- The problem statement should be precise, clear, and free from ambiguity.

}

Problem Analysis {

- Problem analysis is the process of breaking the identified problem into smaller, manageable components so that a logical solution can be designed.
- This task is further subdivided into six subtasks namely:
 - a. Specifying the Objective
 - b. Specifying the Output
 - c. Specifying Input Requirements
 - d. Specifying Processing Requirements
 - e. Evaluating the Feasibility
 - f. Problem Analysis Documentation

}

Problem Analysis {

Specifying the Objective :

- To solve a problem effectively, it's essential to clearly define it.
- Simple problems can be stated easily, allowing a quick transition to program design.
- However, complex problems involving multiple subsystems require thorough analysis, extensive research, and careful coordination of people, processes, and programs.

}

Problem Analysis {

Specifying the Output :

- To determine system inputs, first identify the desired outputs.
- Designing output forms and formats helps specify results clearly. End users, who benefit from the software, are best suited to evaluate these forms.
- Programmers can create various designs, which must be assessed for usefulness.

}

Problem Analysis {

Specifying Input Requirements :

- Once outputs are defined, the necessary inputs and data sources must be identified.
- For instance, in a student record program, inputs like name, address, and roll number may come from the students or their supervisor.

}

Problem Analysis {

Specifying Processing Requirements :

- After defining outputs and inputs, the process for converting inputs into outputs must be specified.
- For replacing or supplementing an existing system, evaluate current procedures for potential improvements.
- If creating a new system, study similar systems to guide development.

}

Problem Analysis {

Evaluating the Feasibility :

- After completing the initial steps, evaluate the practicality and feasibility of the solution.
- For replacing an existing system, assess whether the proposed improvements outperform the current system or similar alternatives.

}

Problem Analysis {

Problem Analysis Documentation :

- At the end of program analysis, document all progress, including program objectives, output and input specifications, processing requirements, and feasibility, to ensure clarity and reference for further development.

}

Program Design {

- program design, involves creating algorithms, flowcharts, and pseudo code to structure the program for user-friendliness, feasibility, and optimization. Using the top-down design approach, tasks are divided into smaller modules and sub-modules, simplifying the programming process into manageable, independent parts.
- In program design we are mainly interested in designing:
 - Algorithms
 - Flowcharts
 - Pseudo codes

}

Program Design {

Algorithm

- An algorithm is an effective step-by-step procedure for solving a problem.
- It is required to decide a solution before writing a program. The procedure of representing the solution in a natural language called an algorithm.
- The same problem can be solved with different methods.
- It captures and refines all the aspects of the desired solution.

}

Program Design {

Flowchart

- Flowchart is basically a pictorial or diagrammatic representation of an algorithm using standard symbols.
- Flowchart is a graphical representation that explains the sequence of operations to be performed in order to solve a problem under consideration.

}

Program Design {

Pseudo codes

- Pseudo code is structured English for describing algorithms concisely.
- As the name suggests, pseudo code does not obey the syntax rules of any particular programming language
- It is not a real programming code. It allows the designer to focus on main logic without being distracted by programming languages syntax.
- Pseudo code is the intermediate state between an idea and its implementation(code) in a high-level language.

}

Coding {

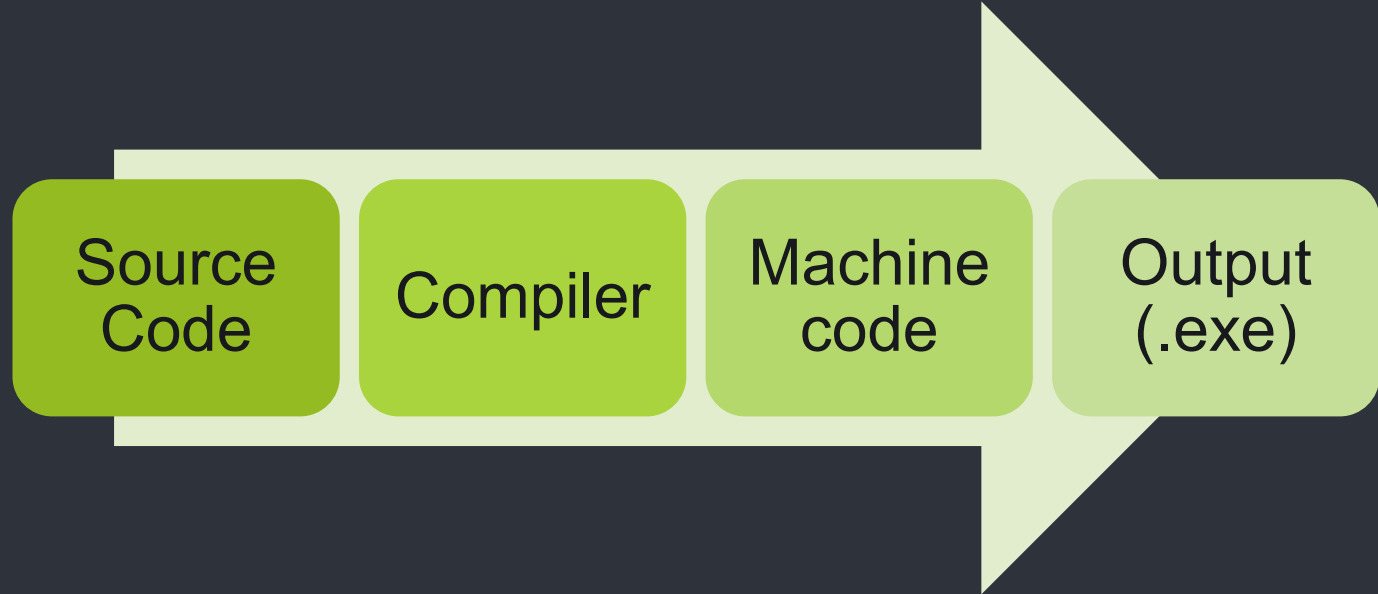
- In this stage, process of writing actual program takes place.
- A coded program is most popularly referred to as a source code.
- The coding process can be done in any language (high level and low level).
- The actual use of computer takes place in this stage in which the programmer writes a sequence of instructions ready for execution.
- Coding is also known as programming.

Compilation and Execution {

- Compilation is the process of converting the source code written by a programmer into machine code or an intermediate form that the computer's processor can understand and execute. This process is carried out by a compiler.
- Execution is the process of running the compiled code (or interpreted code) on a computer. It refers to the actual process of carrying out the instructions defined in your program.

}

Compilation and Execution {



}

Debugging and Testing {

- The designed and developed program undergoes several rigorous tests based on various real-time parameters and the program undergoes various levels of simulations.
- It must meet the user's requirements, which have to respond with the required time. It should generate all expected outputs to all the possible inputs.
- Industries follow some testing methods like system testing, component testing and acceptance testing while developing complex applications.
- The errors identified while testing are debugged or rectified and tested again until all errors are removed from the program.

}

Debugging and Testing {

Error

- An error in programming refers to any issue or problem that prevents a program from functioning as expected.
- Errors can occur during different stages of development, such as during writing, compiling, or running the program.
- They can range from simple syntax mistakes to complex logic errors that produce incorrect results.

}

Debugging and Testing {

Types of Error:

Syntax Errors:

These occur when the code violates the grammatical rules of the programming language.

Logical Errors:

These are bugs that cause the program to run but produce incorrect or unintended results. The program executes without crashing, but the output is wrong.

Runtime Errors:

These occur while the program is running, causing the program to crash or behave unexpectedly.

}

{ Thank You }